Comprehensive HLD Specification: Component Diagrams, DFDs (Level 0–2), API Contracts & Capacity Models

Governing Authority: Ministry of Cooperation, Government of India • *Theme: Sahakar Se Samriddhi*
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Architectural Scope Definition: In strict adherence to enterprise software engineering standards, this High-Level Design (HLD) covers the four macro-architectural pillars: **(1) Component Diagrams** defining major modular boundaries and subsystems; **(2) Data Flow Diagrams (DFDs)** detailing Level 0, Level 1, and Level 2 process flows; **(3) High-Level API Specifications** establishing inter-service interface contracts; and **(4) Quantitative Capacity Models** estimating peak QPS, network bandwidth, 5-year database storage, Redis in-memory footprints, and compute cluster sizing.

1. Component Diagrams & Subsystem Boundaries

Seva Setu is decomposed into eight enterprise components, isolating high-frequency geospatial ingestion from financial and governance pipelines. The architecture enforces clean separation of concerns and loose coupling.

Subsystem Component	Internal Modules	Collaborating Components & External Interfaces
Ingress & Edge Component	Cloudflare Edge CDN, Nginx/Traefik Reverse Proxy, FastAPI ASGI Gateway.	Collaborates with all downstream services. Handles SSL termination, rate-limiting, and JWT RBAC validation.
Identity & Verification Component	Digilocker Adapter, Gemini 1.5 Flash OCR Parser, e-Shram Tokenizer.	Interfaces with UIDAI, e-Shram, and S3 Encrypted Media Vault. Supplies verified worker badges to Spatial Engine.
Geospatial Triage & Dispatch	PostGIS Spatial Query Engine, Multi-Objective Ranking Math, Redis GEOSEARCH.	Consumes worker GPS stream from Redis. Dispatches bookings to Worker Radar via WebSocket push hub.
Fallback & Continuity Coordinator	Rejection Ledger, En-Route Hurdle Resolver, Escalation State Machine.	Listens to worker hurdle webhooks. Re-escalates bookings to next nearest guild peer without terminating session.
Decoupled 4-Layer Pricing	Unbundled Fare Calculator, Seva Subsidy Manager, Dynamic Quota Guard.	Computes transparent line items. Verifies NGO Darpan IDs, checks district pool liquidity, and executes split payouts.
Material Procurement B2B	Pre-Authorization Engine, Cooperative Hardware QR Pass, POS Webhook Listener.	Interfaces with Citizen App and Cooperative Hardware Merchants. Captures 4% B2B wholesale trade margins.
Safety Command & SOS Component	Active Telemetry Compiler, Telephony Bridge, Police 112 ERSS Client.	Triggered by Citizen/Worker SOS. Transmits encrypted GPS, Aadhaar hash, and vehicle details to Police 112.
Ministry Analytics Component	District Deficit Aggregator, PACS Training Planner, Macro Revenue Auditor.	Pulls completed booking logs from PostgreSQL. Generates district deficit heatmaps for Ministry officials.

2. Data Flow Diagrams (DFD Level 0, Level 1 & Level 2)

A. DFD Level 0: Enterprise Context Data Flow

• **Citizen Flow:** Ingests service category, sub-scope, location, and urgency tier $ ightarrow$ Receives unbundled invoice, live worker coordinates, and START_OTP.
• **Worker Guild Flow:** Ingests live GPS telemetry and radar accept/decline $ ightarrow$ Emits en-route hurdle cancellation or on-site spare part quotes $ ightarrow$ Receives 100% direct labor payout.
• **PACS / Cooperative Flow:** Performs physical tool vetting and issues credential seals $ ightarrow$ Receives 1.5% statutory welfare cess (PMSBY insurance fund).
• **Ministry Flow:** Ingests district deficit heatmaps and audit ledgers $ ightarrow$ Emits targeted PACS vocational training deployment orders.
• **National DPI Flow:** Transmits encrypted Aadhaar tokens and live emergency telemetry $ ightarrow$ Receives UIDAI verification and Police 112 ERSS dispatch confirmation.

B. DFD Level 1: Subsystem Process Decomposition

Decomposes data transformations across seven primary persistence stores: **D1: Users & Worker Guild Vault** (Credentials, Float Balance); **D2: PostGIS Spatial Coordinates** (Active Points EPSG:4326); **D3: Bookings & Transparent Ledgers** (Invoices, States); **D4: B2B Material Procurements** (Parts Quotes, Merchant Invoices); **D5: Seva Subsidy Reserve Pool** (Surpluses, CSR Vouchers); **D6: Safety & Police Incident Log** (Encrypted Telemetry); **D7: National District Deficit Ledger** (Trade Demand Indices).

C. DFD Level 2: Service Booking, Fallback & Material Pass Execution Flow

1. **Quote & Subsidy Check:** Citizen request triggers Process 2.1 (Sub-scope parse) $ ightarrow$ Process 2.4 checks Darpan ID and validates the Subsidy Guardrail against D5 (Liquidity Check) $ ightarrow$ Persists unbundled invoice to D3.
2. **Spatial Matching & WebSocket Dispatch:** Process 3.1 executes PostGIS ST_DWithin on D2 $ ightarrow$ Computes Triage Ranking $ ightarrow$ Pushes broadcast to Candidate #1.
3. **Zero-Downtime Fallback:** If Worker #1 declines or triggers an en-route hurdle, Process 3.6 logs the rejection in D3 and instantly re-escalates to Candidate #2 without terminating the booking.
4. **Dynamic Material Procurement Pass:** On-site parts request triggers Process 4.4 (Citizen pre-auth) $ ightarrow$ Generates QR Pass $ ightarrow$ Partner store POS webhook confirms pickup in D4.
5. **Dual OTP Settlement:** Process 5.2 validates END_OTP $ ightarrow$ Transfers 100% labor to worker bank, 1.5% to PACS PMSBY fund, and streams coordinates to D7 (Ministry Deficit Heatmap).

3. High-Level API Specifications & Interface Contracts

Core Enterprise Interface Signatures:

```
// 1. ORDER DISPATCH & QUOTE INTERFACE (POST /api/v1/orders/quote-and-dispatch)
REQUEST: {
  "service_category": "ELECTRICIAN", "sub_scope_code": "CIRCUIT_BREAKER_TRIP_FIX",
  "urgency_tier": "P1_CRITICAL", "pricing_horizon": "HOURLY", "payment_mode": "UPI_ESCROW",
  "customer_coords": { "lat": 28.6315, "lng": 77.2167 }, "is_ngo": true, "darpan_id": "DL/2021/00892"
}
RESPONSE (201 Created): {
  "order_id": "ord_9b1deb4d", "order_ref": "SEVA-DEL-2026-9041", "status": "BROADCASTING",
  "start_otp": "8241", "end_otp": "5913",
  "invoice": { "base_labor": 240.00, "transit": 30.00, "emergency_surcharge": 120.00,
               "platform_fee": 8.40, "cess": 3.60, "ngo_discount": -94.80, "total_payable": 379.20 }
}

// 2. EN-ROUTE HURDLE FALLBACK INTERFACE (POST /api/v1/dispatch/{order_id}/report-hurdle)
REQUEST:  { "worker_id": "wkr_101", "hurdle_type": "VEHICLE_PUNCTURE", "coords": { "lat": 28.6322, "lng": 77.2210 } }
RESPONSE: { "hurdle_logged": true, "fallback_status": "RE_ESCALATED", "reassigned_worker_id": "wkr_106_amit" }

// 3. ON-SITE MATERIAL PRE-AUTH INTERFACE (POST /api/v1/procurement/{order_id}/pre-authorize)
REQUEST:  { "item_id": "mat_4892", "amount": 450.00, "signature": "sig_escrow_preauth_8829" }
RESPONSE: { "status": "PRE_AUTHORIZED", "qr_pass": "https://vault.sevasetu.gov.in/qr/pass_9921.png", "b2b_margin": 18.00 }
```

4. Quantitative Capacity Models & Sizing Computations

Rigorous back-of-the-envelope calculations model system requirements for nationwide deployment across 500,000 certified technicians:

Metric Dimension	Formula & Sizing Derivation	Provisioned Engineering Specification
Peak Booking QPS	(1,000,000 daily bookings / 86,400s) × 3.0 diurnal peak factor.	**35 Bookings / Sec** (Readily handled by ASGI event loop).
Peak GPS Telemetry QPS	(200,000 concurrent workers / 10s ping interval) × 2.5 emergency surge.	**50,000 GPS Pings / Sec** (Ingested into Redis Streams).
Network Bandwidth	GPS (50k × 250B = 100 Mbps) + Trans API (2k × 2.5KB = 40 Mbps).	**140 Mbps Continuous Peak** (1 Gbps redundant edge pipe).
5-Year PostgreSQL Storage	1M bookings/day × 1.5 KB × 365 days × 1.5 index multiplier × 5 yrs.	**2.75 Terabytes** (Hot active DB with archiving >18 months).
5-Year Media Storage (S3)	1M jobs/day × 2 photos × 200 KB × 365 days × 5 years.	**730 Terabytes** (Hot S3 90 days → S3 Glacier Lifecycle).
Redis In-Memory RAM	Worker Geopoints (12.8MB) + Sessions (75MB) + Tokens (1GB) × 4.0 safety.	**16.0 Gigabytes** (3-node Redis cluster on cache.t4g.medium).
Compute Cluster Allocation	50,000 GPS QPS (42 cores) + 35 Trans QPS (16 cores) @ 65% target CPU.	**20 ASGI Nodes** (4 vCPU / 16 GB RAM each with K8s HPA).

High-Level Design Certification: This HLD Engineering Blueprint establishes the macro-systemic feasibility of Seva Setu. By mathematically modeling capacity across 50,000 peak GPS QPS, separating concerns into 8 decoupled components, defining formal DFD Level 0–2 flows, and specifying high-level API boundary contracts, it guarantees enterprise stability for nationwide public service deployment.